FINAL PROJECT PROPOSAL

Higher Diploma in Software Engineering – Batch 17

ICBT Campus

OptoPick

AI-Powered Intelligent Optical Center Management System

A Smarter Approach to Optimal Selection

Item	Details
Project Type	Final Semester Project
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1. Introduction

This project proposal is submitted to fulfil the software development requirements of the Group Computing Project module. The purpose of this document is to present a structured plan for the proposed project, demonstrating its feasibility, development approach, and expected outcomes in accordance with academic and professional standards.

This proposal provides a comprehensive overview of the proposed study by presenting the background and motivation, client information, current business environment, problem analysis, project aim and objectives, proposed solution, project scope, feasibility analysis, development methodology, work breakdown structure, project schedule, and resource requirements. Each section is organized to provide a clear understanding of the project from its initial concept to its planned implementation.

Overall, this document serves as a guide for supervisors, evaluators, and other stakeholders by presenting the project in a logical and systematic manner while providing the information required to assess the proposed work before development begins

2. Background and Motivation

2.1 Client Background

The client organization for this project is Indian Optician, an independent optical retail and eye-care business operating in Nintavur, in the Ampara District of Sri Lanka's Eastern Province. The center provides eye examinations, prescription eyewear, frames, lenses and related accessories to customers from Nintavur and the surrounding communities.

Item	Details
Client Organization	Indian Optician
Owner	Mr. Musthaq
Address	No 67/1, Main Street Nintavur - 13
Year of Establishment	2024
Organisation Type	Independent optical retail and eye-care center

Table 1: Client organization profile



Figure 1: Indian Optician – Client Premises

The figure will show the client premises in Nintavur and will be supplied by the client during the requirement-gathering stage.

2.2 Company History

Indian Optician was established in Nintavur by its owner, an eye consultant, to provide accessible professional eye-care and eyewear services to the local community without customers having to travel to larger towns. Since its establishment 2024, the center has developed a customer base that relies on it for routine eye examinations, spectacle prescriptions and the replacement of frames and lenses.

The center has grown through personal service and word-of-mouth recommendation. As in many small optical practices in regional Sri Lanka, its administrative practices developed informally over time. Eye-power readings are written into a marking book, customer orders are written by hand and partially re-entered into Microsoft Excel files, and daily sales and accounts are maintained by the owner in separate monthly spreadsheets. Completed frame and prescription details are sent to an external fitting technical center, which prepares and fits the lenses before the finished spectacles are returned for collection. Through this service Indian Optician gained customer trust by providing spectacles and related items in reasonable price.

2.3 Vision

To be the most trusted optical center in the Nintavur region by combining professional eye care with modern, reliable and professional service that make every customer's experience convenient and personal.

2.4 Mission

Indian Optician is committed to providing accurate eye examinations, suitable prescription eyewear and honest professional advice at fair prices. The center aims to maintain complete and secure customer records, to manage its products and suppliers efficiently, and to build long-term relationships with customers through timely follow-up and consistent service quality, supported by appropriate use of technology.

2.5 Roles and Responsibilities of the Client Organization

Indian Optician operates with a compact two-person structure. Both roles will participate in the project as key stakeholders by providing domain knowledge, validating requirements, reviewing designs and evaluating the completed system. The organizational roles described below also form the basis of the two user roles in OptoPick: Administrator / Optometrist and Optical Staff.

2.5.1 Owner / Eye Consultant (Optician)

The owner is the principal decision-maker and the sole professional responsible for eye examinations. In the context of the project, the owner acts as the Product Owner and the primary source of business and clinical requirements.

1. Conducts eye examinations and determines each customer's correctable eye power (SPH, CYL, AXIS, ADD and PD).

2. Records examination results and prescriptions, currently in a marking book.
3. Provides professional advice on suitable lens types, materials and coatings.
4. Reviews daily sales, revenue and accounts maintained in Microsoft Excel, with monthly records saved in separate files.
5. Makes purchasing and replenishment decisions for frames, lenses and accessories.
6. Maintains relationships with suppliers and the external fitting technical center.
7. Defines business rules such as pricing, discounts and follow-up periods.
8. Reviews and approves project deliverables and participates in user acceptance testing.

2.5.2 Optical Staff Member

The optical staff member handles front-of-shop operations and day-to-day administration. In the project, this role represents the primary operational user and provides detailed knowledge of the sales, ordering and stock workflows.

1. Welcomes customers visiting the center and assists them in selecting frames.
2. Explains frame and lens options and finalizes frame and lens type negotiations with customers.
3. Handles customer orders manually and records customers' vision details on paper and in an Excel file.
4. Cleans the shop and refills frames on their corresponding shelves.
5. Sends confirmation photographs of selected frames together with the customer's prescription to the fitting technical center.
6. Dispatches order parcels to the fitting technical center and handles collection of completed spectacles.
7. Provides operational feedback during requirement validation and testing.

2.5.3 Client Responsibilities towards the Project

- Providing access to current business processes, sample (anonymized) records and product information.
- Participating in interviews, observation sessions and requirement-validation meetings.
- Reviewing user-interface prototypes and providing feedback during sprint reviews.
- Defining operational parameters such as low-stock thresholds and customer follow-up periods.
- Participating in user acceptance testing and evaluating the usefulness of the AI features.

2.6 Current Business Process

2.6.1 Overview of the Current Process

At present, the operations of Indian Optician are carried out through a combination of paper records, verbal communication and separate Excel files. There is no single system that connects customer details, eye-examination results, product information, stock levels and sales. The main activities of the current process are described below.

2.6.2 Customer Reception and Eye Examination

- The optical staff member welcomes the customer and records basic details on paper.
- The owner examines the customer's eyes and writes the correctable eye power into a marking book.
- The written prescription is handed physically to the staff member for the dispensing stage.
- When a customer returns in later years, previous readings must be searched for manually in the marking book or Excel files.

2.6.3 Frame and Lens Selection and Ordering

- The customer browses the frames displayed on the shelves with assistance from the staff member.
- Lens type, material and coating are discussed and agreed based mainly on staff experience and the owner's advice.
- The order, including the frame and vision details, is written manually and later re-entered into an Excel file.
- A confirmation photograph of the frame and the prescription is sent to the fitting technical center, followed by the physical parcel.

2.6.4 Billing and Payment

- Order totals and discounts are calculated manually.
- Payments, advances and balances are recorded on paper or in a spreadsheet.
- The owner reviews daily sales and revenue in Excel and stores each month's records in a separate file.

2.6.5 Stock Management

- Frames and lenses are restocked based on visual inspection of the shelves and periodic manual counts.
- Stock records are not updated automatically when a sale is made.
- Purchasing decisions are based on the owner's judgement rather than on analyzed sales history.

2.6.6 Customer Follow-up

- There is no systematic mechanism to identify customers who are due for an eye review or eyewear replacement.
- Follow-up communication, where it occurs, depends on staff memory and individual initiative.

2.6.7 Challenges in the Current Process

Challenge	Description	Impact
Fragmented records	Customer details and prescriptions are spread across a marking book, paper orders and multiple Excel files.	Lost or illegible records, duplicated effort and slow retrieval.
Manual re-entry	The same information is written by hand and later typed into spreadsheets.	Transcription errors and additional administrative workload.
Manual billing	Totals, discounts and balances are calculated by hand.	Slower checkout, arithmetic errors and unrecorded payments.
Disconnected stock	Sales do not update stock records.	Inaccurate stock levels and attempts to sell unavailable items.
Intuition-based purchasing	Replenishment relies on observation rather than data.	Stock shortages of popular items and capital tied up in slow-moving items.
No structured follow-up	No record-driven reminders for reviews or replacements.	Lost repeat business and weaker customer relationships.
Inconsistent lens advice	Lens recommendations depend on individual experience.	Variable dispensing quality and customer experience.

Table 2: Challenges in the current business process

2.7 Motivation

2.7.1 Overview of Existing Optical Center Management Systems

Several optical practice management systems are used around the world. However, most are designed for large practices in the United States, the United Kingdom or India. They provide limited support for a small, independent optical center in Sri Lanka such as Indian Optician. The following analysis examines the main systems and their limitations.

2.7.2 Crystal Practice Management (Crystal PM)

Platform Description:

Crystal PM is optometry practice management software developed by Crystal Practice Management, Austin, Texas, USA. It is offered both cloud-hosted and installed in the office (Crystal Practice Management, nd).

Main Features:

- Custom electronic health records and templates
- E-Prescribing
- Frame, contact lens and lens inventory, with lab ordering
- Integrated payments and insurance claims processing
- Two-way texting and notifications when products are ready for pick-up
- Online scheduling and online forms
- Basic and enhanced reporting

Limitations for Indian Optician:

- Designed for the US healthcare and insurance environment
- Tiered subscription pricing, with base prices available only on request
- No AI-assisted lens decision support
- No AI-based inventory demand insight
- No AI-based customer follow-up intelligence

Critical Evaluation:

Crystal PM is a complete system for US optometry practices. However, its certified health records, insurance claims and subscription model are unnecessary and costly for a two-person optical center in Nintavur. It also does not provide intelligent decision support for lens selection, stock planning or customer retention.

2.7.3 Acuitas 3 (Ocuco)

Platform Description:

Acuitas 3 is a cloud-based practice management system for independent optical practices, developed by Ocuco Ltd, Dublin, Ireland (Ocuco, n.d).

Main Features:

- In-store and online appointment booking
- Clinical records and configurable eye-examination workflows
- Validated dispensing to reduce remakes

- Real-time stock management
- Integrated card payments (OcuPay)
- Automated recalls and multi-channel marketing
- Power BI reporting and analytics
- NHS General Ophthalmic Services (GOS) form management

Limitations for Indian Optician:

- Designed for the UK and Irish market and the NHS environment
- Subscription-based pricing suited to larger independents and chains
- No AI-assisted lens decision support
- No generative-AI explanations for stock or customer insights
- More complex than a two-person center needs

Critical Evaluation:

Acuitas 3 shows the value of automated recalls and real-time stock control, both of which OptoPick also provides. However, its NHS-oriented features, cost and complexity make it unsuitable for a small Sri Lankan optical center. It also offers no explainable AI support for daily decisions.

2.7.4 RevolutionEHR

Platform Description:

RevolutionEHR is a cloud-based electronic health record and practice management platform for optometry practices in the United States. It is used by more than 13,000 eye-care professionals (RevolutionEHR, n.d).

Main Features:

- Electronic health records and examination workflows
- Patient dashboard showing exam history, balance and insurance
- Inventory management
- Insurance claims, insurance verification and payments (RevClear, RevCheck, RevPay)
- Reminders, campaigns and online booking (RevEngage)
- AI Scribe for spoken clinical note-taking

Limitations for Indian Optician:

- Designed for US insurance and regulatory requirements
- Monthly subscription charged per doctor, plus one-time conversion fees

- AI is limited to clinical documentation (AI Scribe)
- No AI-assisted lens decision support
- No AI-based inventory demand or customer follow-up intelligence

Critical Evaluation:

RevolutionEHR is the only system reviewed that uses AI, but only to speed up clinical note-taking. It does not apply AI to lens selection, stock planning or customer retention, which are the main needs of Indian Optician.

2.7.5 Vyapar Optical Shop Billing Software

Platform Description:

Vyapar is a billing and accounting application developed by Simply Vyapar Apps Pvt. Ltd., Bengaluru, India, and marketed to optical shops among other retailers. It runs on Windows, macOS, Android and iOS, with offline use and cloud sync (Vyapar, n.d).

Main Features:

- Customer records with order history
- Prescription storage and expiry reminders
- Frame, lens and accessory inventory with low-stock alerts
- GST billing, barcode scanning and multiple payment methods
- WhatsApp transaction messages and loyalty points
- Financial reports

Limitations for Indian Optician:

- Mainly a general retail billing and accounting tool
- Website does not describe structured eye-examination records (SPH, CYL, AXIS, ADD, PD) or chronological prescription history
- Website does not describe role-based clinical access
- No AI-powered decision support
- Designed around the Indian GST tax system

Critical Evaluation:

Vyapar is affordable and practical for billing in South Asian retail shops. However, it is not designed around optical clinical workflows and offers no intelligent support for lens selection, demand analysis or customer retention.

2.7.6 Gap Analysis

Platform	Structured Prescription History	Inventory and Low-Stock Control	Recall / Follow-up	AI Features
Crystal PM	Yes	Yes	Yes	No
Acuitas 3	Yes	Yes	Yes	No
RevolutionEHR	Yes	Yes	Yes	Limited (AI Scribe only)
Vyapar	Partial	Yes	Partial	No
OptoPick (Proposed)	Yes	Yes	Yes (reviewed by staff)	Yes (LensIQ, OptoForecast, OptoRecall)

Table 3: Gap analysis of existing alternatives

2.7.7 Identified Gaps in Existing Systems

1. **Regional Gap:** None of the systems reviewed is designed for small independent optical center in Sri Lanka.
2. **Cost Gap:** Most systems use subscription pricing designed for larger practices, which is disproportionate for a two-person center.
3. **AI Decision-Support Gap:** No system provides AI-assisted lens recommendations based on prescription, lifestyle and available stock.
4. **Inventory Intelligence Gap:** No system interprets sales trends with AI to give explainable demand and stock-risk insights.
5. **Customer Retention Gap:** Existing recall features are rule-based only and do not explain why a customer may need attention or draft reviewable messages.
6. **Integration Gap:** No affordable system combines optical clinical records, sales-linked inventory and AI decision support in one platform.

3. Problem in Brief

Optical centers combine two kinds of work: a clinical service, in which eye examinations produce prescriptions that must be recorded accurately and retained over many years, and a retail operation, in which frames, lenses and accessories must be stocked, sold and replenished efficiently. At Indian Optician, both kinds of work are currently supported by paper records and disconnected spreadsheets. Through interviews with the owner and staff member and an examination of the current workflow, the following nine problems have been identified. For each problem, the issue, its organizational impact, the inadequacy of the current approach and the relevant OptoPick response are discussed. Illustrative scenarios are included to show how each problem may affect the center; these are realistic hypothetical examples and are not presented as recorded historical incidents.

Problem 1: Fragmented Customer and Prescription Records

Problem Definition:

Staff manually write customer details and prescription values onto paper cards. This creates a fragmented workflow where physical files are easily misplaced, leading to illegible data, manual transcription errors, and severe data retrieval issues.

Evidence:

A review of 25 customer records at Indian Optician (2026) found that 6 records existed in more than one place. In 4 records, the prescription values did not match between copies or could not be read. Research also shows that human error is unavoidable in manual record-keeping and that electronic records help reduce errors (Chimbo, B & Motsi, L., 2024).

Environmental Factors Contributing to the Problem

- Reliance on handwritten records and a marking book.
- The same data entered twice, on paper and in Excel.
- No standard format for recording prescriptions.
- Limited staff time during busy periods.

Problem 2: Difficult to Find Eye Examination Histories

Problem Definition:

When a customer returns after several years, finding their previous eye-examination history in old marking books and local Excel files takes a long time. This stops the optometrist from easily comparing past and present prescriptions to track changes in vision.

Evidence:

In a timed exercise at Indian Optician (2026), finding the previous examination record of 10 returning customers took an average of 6-10 minutes, and 3 records could not be found. A study of electronic medical record (EMR) implementation reported that

average documentation time fell from 120 minutes to 45 minutes after moving from paper to electronic records (Torab-Miandoab, A, et al., 2025).

Environmental Factors Contributing to the Problem

- Records spread across several years of marking books.
- Monthly data stored in separate Excel files.
- No index or search by customer.
- No chronological view of a customer's prescriptions.

Problem 3: Time-Consuming Order and Billing Processes

Problem Definition:

Order totals, promotional discounts and invoices are calculated and written by hand. This slows checkout and leads to arithmetic errors and unrecorded payments.

Evidence:

A comparison of 80 paper bills with the monthly Excel sales file at Indian Optician (2026) found 5 bills with calculation errors and 3 advance or balance payments that were not recorded. Manual billing took an average of 9 minutes per order. Retail sources report that manual point-of-sale calculation contributes to revenue leakage of about 3–5% (DohAssist, 2026).

Environmental Factors Contributing to the Problem

- Orders that combine frames, lenses, coatings, discounts and advances.
- Manual calculation without billing software.
- Payments recorded separately from orders.
- Pressure to serve customers quickly.

Problem 4: Lack of Centralized Departmental Communication

Problem Definition:

The examination room, sales floor and storeroom work independently. Prescriptions written by the optometrist are handed physically to the staff member, which risks miscommunication, lost hand-offs and dispensing errors.

Evidence:

The owner of Indian Optician reported 3 cases in the past 6 months where a prescription slip was misplaced, misread or attached to the wrong order before being sent to the fitting technical center. Healthcare evidence shows that paper hand-offs increase the risk of errors from misread handwriting or lost documents (Institute of Medicine, 2007). **Environmental Factors Contributing to the Problem**

- Handwritten prescription slips passed between people.
- No shared digital record for owner and staff.
- Photos and parcels sent manually to the fitting technical center.
- Several customers handled at the same time.

Problem 5: Inaccurate Stock Levels for Frames and Lenses

Problem Definition:

The center relies on periodic manual stock counts. Staff sometimes try to sell frames or lenses that are no longer in stock, which leads to cancelled orders, delays and frustrated customers.

Evidence:

A physical count of 30 frame models at Indian Optician (2026) found that 5-8 models had quantities different from the recorded stock. The owner estimated 2-4 orders per month are delayed or changed because a selected item was unavailable. Small-business sources report that up to 43% of small businesses do not track inventory accurately (PendaStock, 2026).

Environmental Factors Contributing to the Problem

- Sales not linked to stock records.
- Manual and occasional stock counting.
- Large variety of frame models, colours and lens types.
- No low-stock alerts.

Problem 6: Lack of Predictive Product Demand

Problem Definition:

Management purchases new inventory based on guesswork rather than historical sales data. This lack of predictive insight results in the overstocking of unpopular items and capital wasted on slow-moving inventory

Evidence:

At Indian Optician (2026), About 10–15% of models with no sale in 6 months, tying up roughly LKR 50,000–150,000; 3–5 popular models ran out at least once

Environmental Factors Contributing to the Problem

- Sales data stored in separate monthly Excel files.
- No analysis of sales by product or over time.
- Purchasing based on memory and experience.
- Seasonal demand changes, such as festivals and school terms.

Problem 7: Poor Post-Sale Tracking and Customer Retention

Problem Definition:

Purchase dates and order status are kept in paper logbooks. As a result, staff do not follow up on uncollected orders or remind customers when their annual eye check-up is due.

Evidence:

At Indian Optician (2026), 4-8 completed spectacle orders remained uncollected for more than 2 weeks. The center currently sends no reminders for annual check-ups. Optometry practice sources indicate that practices can lose up to 50% of their patients over five years without active recall strategies (Jelo, 2026).

Environmental Factors Contributing to the Problem

- Order status recorded only on paper.
- No reminder system.
- Follow-up depends on staff memory.
- Limited staff time.

Problem 8: Inconsistent Lens Recommendations

Problem Definition:

Lens selection relies mainly on individual staff experience rather than a standard method. This makes it hard to recommend suitable lenses consistently based on a customer's prescription and lifestyle.

Evidence:

In an interview, the owner of Indian Optician stated that "lens advice depends mainly on the staff member's experience and is not recorded". In an illustrative scenario, if the owner and the staff member were each given the same 5 sample customer cases, their lens recommendations could differ in 3 of the 5 cases. Ophthalmic dispensing literature emphasizes that lens selection should consider prescription, material, index, coating and the wearer's lifestyle (Brooks, C.W & Borish, I.M., 2007).

Environmental Factors Contributing to the Problem

- No structured lens-selection guideline.
- Growing range of lens materials and coatings.
- Lifestyle factors not recorded.
- Different levels of experience between staff.

Problem 9: Poor Patient Follow-up and Limited Customer Retention

Problem Definition:

The business lacks an automated system to track the patient eyewear lifecycle or prompt necessary reviews, resulting in poor follow-up that severely limits long-term engagement and allows customers to drift to competitors.

Evidence:

In Indian Optician’s Excel files for 2025, 35-50% have not returned in more than two years, and none received a follow-up message. Optometry practice sources report that winning a new patient costs six to seven times more than keeping an existing one.

Environmental Factors Contributing to the Problem

- Customer history not consolidated.
- No tracking of customers who have stopped visiting.
- Competing optical shops in nearby towns.
- No organized follow-up communication.

Summary of Problems

Problem	Key Impact	Affected Stakeholders	OptoPick Response
1. Fragmented customer and prescription records	Transcription errors, lost records	Owner, staff, customers	Centralized customer and prescription records
2. Difficult to find eye examination histories	Delayed consultations, weak comparison	Owner, customers	Chronological prescription history
3. Time-consuming order and billing	Slow checkout, revenue leakage	Staff, owner, customers	Automated sales and billing
4. Lack of centralized departmental communication	Lost hand-offs, dispensing errors	Owner, staff	Connected digital workflow
5. Inaccurate stock levels	Cancelled orders, lost sales	Staff, owner, customers	Sales-linked inventory and alerts
6. Lack of predictive product demand	Stock-outs and overstock	Owner	OptoForecast
7. Poor post-sale tracking and retention	Uncollected orders, missed reviews	Staff, customers	OptoRecall and communication
8. Inconsistent lens recommendations	Variable dispensing quality	Owner, staff, customers	LensIQ
9. Poor patient follow-up and limited retention	Customer attrition	Owner, customers	OptoRecall

Table 4: Summary of identified problems

4. Aim

The aim of this project is to design and develop an intelligent optical center management system that digitizes manual workflows, centralizes patient records, facilitates AI-driven lens selection and automates customer retention.

5. Objectives

The following objectives have been defined to achieve the project aim:

Objective 1: To develop a centralized digital database that securely stores customer profiles and prescriptions, eliminating paper-based transcription errors and allowing instant data retrieval.

Objective 2: To implement a digital chronological prescription history feature that enables optometrists to instantly retrieve and compare a patient's historical and current eye-examination records on a single screen.

Objective 3: To integrate an automated billing and payment management module that instantly calculates product prices, applies discounts accurately and generates digital invoices to streamline the checkout workflow.

Objective 4: To create a connected operational workflow that seamlessly transfers digital prescriptions from the examination room directly to the sales and dispensing dashboard, ensuring zero data loss between departments.

Objective 5: To deploy a real-time inventory management system that automatically updates stock quantities of frames and lenses the moment a sale or order is processed.

Objective 6: To implement a predictive inventory management module (OptoForecast) that analyses historical sales data to forecast future product demand and prevent stock shortages.

Objective 7: To build an AI-powered customer lifecycle tool (OptoRecall) that monitors purchase histories to automatically trigger order-ready notifications and future eyewear review reminders, improving patient retention.

Objective 8: To build an AI-assisted recommendation engine (LensIQ) that standardizes lens selection by evaluating precise clinical data and lifestyle inputs.

Objective 9: To create an intelligent customer lifecycle tool (OptoRecall) that automatically identifies and re-engages lapsed patients through targeted follow-up reminders.

Objective	Problem(s) Addressed	OptoPick Component
Objective 1	Problem 1	Customer Management; Digital Prescription Records
Objective 2	Problem 2	Prescription History
Objective 3	Problem 3	Sales and Billing
Objective 4	Problem 4	Shared customer profile and sales workflow
Objective 5	Problem 5	Inventory and Stock-Movement Management
Objective 6	Problem 6	OptoForecast
Objective 7	Problem 7	OptoRecall; Customer Communication
Objective 8	Problem 8	LensIQ
Objective 9	Problem 9	OptoRecall

Table 5: Mapping of objectives to problems and system components

6. Proposed Solution

6.1 Overview of the Proposed Solution

6.1.1 Solution Introduction

The proposed solution is OptoPick, an AI-powered intelligent optical center management system developed for Indian Optician, Nintavur. The system replaces the current paper-based and spreadsheet-based processes with a centralized, secure and user-friendly web-based platform. It connects the owner (Administrator / Optometrist) and the optical staff member through a single system that manages customers, eye-examination records, frames, lenses, inventory, sales, suppliers and customer communication.

In addition to these core functions, OptoPick provides three focused artificial intelligence (AI) features, LensIQ, OptoForecast and OptoRecall, which use the Gemini API to support lens selection, stock planning and customer follow-up (Google, n.d). OptoPick remains responsible for all records and calculations, while the AI features only provide suggestions that are reviewed by the user before any decision is made.

6.1.2 Addressing Current Limitations

Currently, the main activities of Indian Optician, including eye-examination recording, customer orders, billing, stock control and customer follow-up, are managed through a marking book, handwritten orders and separate Excel files. These manual methods lead to:

1. Customer details and prescriptions scattered across paper records and spreadsheets.
2. Slow retrieval of previous eye-examination records when customers return.
3. Manual calculation of bills, discounts, advances and balances.
4. Paper hand-offs of prescriptions between the owner and the staff member.
5. Stock records that are not updated when items are sold.
6. Purchasing decisions made without analysis of sales history.
7. Uncollected orders and missed eye-review reminders.
8. Lens recommendations that depend mainly on individual experience.
9. No organized follow-up of customers who stop visiting the center.

6.1.3 Solution Approach

OptoPick replaces these traditional methods with an integrated digital solution combining:

1. A web-based application used on the computer available at the center.
2. A centralized MySQL database that acts as the single source of truth for all records.
3. Role-based access for the Administrator / Optometrist and the Optical Staff.
4. Automatic sales-linked stock updates and low-stock alerts.
5. Artificial intelligence through the Gemini API for lens decision support, demand insight and customer follow-up.
6. Human review of every AI suggestion before it is used.

6.1.4 Justification of the Proposed Solution

The proposed solution has been selected because it directly responds to the problems identified at Indian Optician, suits the size and working style of the center, and improves the existing business process without changing the way the center serves its customers.

Suitability for Indian Optician

Indian Optician is a small, single-branch optical center operated by two people. OptoPick is therefore designed around exactly two user roles that reflect the actual responsibilities of the owner and the staff member. The system runs as a web application on the existing computer at the center, uses free and open-source technologies, and does not require specialized hardware. Its interfaces follow the same order as the current workflow, from customer reception and eye examination to billing and follow-up, so that the staff can adopt it with minimal training. Core functions such as customer records, billing and stock control continue to work even if the AI service is temporarily unavailable.

How the Solution Addresses the Identified Problems

Problem	OptoPick Feature	How It Solves the Problem
1. Fragmented customer and prescription records	Customer Management and Digital Prescription Records	Stores each customer's details and prescriptions once in a centralized database, removing duplicate paper and Excel copies.
2. Difficult to find eye examination histories	Prescription History	Displays all previous examinations in date order so the optometrist can compare past and current values on one screen.
3. Time-consuming order and billing processes	Sales and Billing	Calculates totals, discounts, advances and balances automatically and produces digital receipts and invoices.
4. Lack of centralized departmental communication	Shared customer profile and sales workflow	Makes prescriptions recorded by the optometrist immediately available to the staff member during sales.
5. Inaccurate stock levels for frames and lenses	Inventory Management	Deducts sold items automatically, records every stock movement and shows low-stock and out-of-stock items.
6. Lack of predictive product demand	OptoForecast	Analyzes sales and stock data and highlights rising demand, possible shortages and slow-moving items.
7. Poor post-sale tracking and customer retention	OptoRecall and Customer Communication	Identifies orders ready for collection and customers due for review, and prepares reminders for staff approval.
8. Inconsistent lens recommendations	LensIQ	Suggests suitable lens options from the available catalog based on prescription, lifestyle and budget, with reasons.
9. Poor patient follow-up and limited customer retention	OptoRecall	Highlights lapsed customers, explains why follow-up may be useful and drafts editable messages.

Table 6: How OptoPick addresses the identified problems

Why the Selected Features Are Appropriate

The core modules were selected because each one matches a daily activity observed at the center: customer reception and eye examination, frame and lens selection, ordering and billing, stock control, supplier purchasing and customer follow-up. The three AI features focus on the three decisions that currently depend most on personal experience: choosing a suitable lens, deciding what stock to buy and deciding which customers to contact.

The Gemini API was selected instead of training custom AI models because Indian Optician does not yet have the large volume of historical data needed to train such models, and because a hosted AI service can reason over small, structured summaries prepared by the system at a low cost. Since the center works with eye-care information, every AI result is presented as a suggestion and the final decision always remains with the owner or staff member, which follows recognized guidance on trustworthy AI (National Institute of Standards and Technology (NIST), 2023).

Improvement of the Existing Business Process

Activity	Current Process	With OptoPick
Eye examination	Eye power written in a marking book	Prescription entered once in a structured digital record
Returning customer	Manual search through old books and Excel files	Instant search with full prescription history
Frame and lens selection	Based mainly on experience	Supported by catalog details and LensIQ suggestions
Billing	Manual calculation and handwritten bills	Automatic totals, discounts, balances and digital receipts
Stock control	Visual checks and occasional manual counts	Automatic deduction, stock history and low-stock alerts
Purchasing	Owner’s judgment	Sales reports and OptoForecast demand insights
Customer follow-up	Depends on staff memory	OptoRecall reminders reviewed and sent by staff
Management reporting	Separate monthly Excel files	Dashboard with sales, stock and customer information

Table 7: Comparison of the current process and the proposed process

6.1.5 System Components

The proposed solution consists of four interconnected components:

1. Core Optical Center Management System
2. LensIQ – AI-assisted lens decision support
3. OptoForecast – AI-assisted inventory demand insight
4. OptoRecall – AI-assisted customer follow-up and retention

The Core Optical Center Management System is used daily by both the Administrator /Optometrist and the Optical Staff. LensIQ is available to both roles, while OptoForecast is available to the Administrator / Optometrist and OptoRecall provides full access to the Administrator / Optometrist with restricted follow-up support for the Optical Staff.

6.1.6 System Integration

All components communicate through a Node.js and Express.js backend and a centralized MySQL database. MySQL stores permanent information such as user accounts, customers, prescriptions, frames, lenses, stock quantities, stock movements, sales, payments, suppliers and communication records. When the staff member completes a sale, the sale is recorded, the sold quantities are deducted from stock, a stock movement is recorded and the low-stock status is updated automatically.

The AI features read summarized information from the database and send only the necessary details to the Gemini API through the backend. The results are checked by the system and displayed to the user for review. If the Gemini API is unavailable, the user is informed that AI assistance is temporarily unavailable, while all core functions continue normally.

6.2 Core Optical Center Management System

6.2.1 Overview

The Core Optical Center Management System is the main working environment of OptoPick. It replaces the marking book, handwritten orders and separate Excel files with one secure system that supports the daily work of the owner and the staff member at Indian Optician.

6.2.2 Core Functionality

Authentication and User Access:

- Secure login for the Administrator / Optometrist and the Optical Staff.
- Role-based access to functions and records.
- User account management by the Administrator.
- Password management and protection.

Customer Management:

- Register customers with contact details, age and relevant preferences.

- Search customers quickly by name, telephone number or customer ID.
- View a single customer profile containing purchases, prescriptions and communication history.

Eye Examination and Prescription Records:

- Record SPH, CYL and AXIS for each eye, together with ADD and PD.
- Record examination notes, examination date and the responsible user.
- Keep all previous prescriptions in date order without overwriting them.
- Compare previous and current prescriptions on a single screen.
- Make prescriptions immediately available to the staff member during sales.

Frame and Lens Management:

- Maintain frame details such as brand, model, shape, color, material, size, price and supplier.
- Maintain lens details such as lens ID, type/design, material, refractive index, available coatings, brand, price, supplier and stock/availability status.
- Record the supported SPH, CYL and (for multifocal products) ADD ranges of each lens, based on the specifications supplied by the manufacturer or laboratory, for use in LensIQ compatibility checking.
- Maintain accessory details and prices.

Inventory Management:

- Maintain current stock quantities and low-stock limits.
- Record new stock received from suppliers.
- Deduct sold items automatically when a sale is completed.
- Keep a stock movement history explaining every change in quantity.
- Show In Stock, Low Stock and Out of Stock status.
- Handle authorized returns of products to stock.

Sales and Billing:

- Create sales containing frames, lenses and accessories.
- Calculate totals and discounts automatically.
- Record advances, payments and balances.
- Generate digital receipts and invoices.
- Update the customer’s purchase history automatically.

Supplier Management:

- Maintain supplier details and contact information.
- Record products supplied, purchase records and supplier prices.

Customer Communication:

- Prepare order-ready notifications and eye-review reminders.
- Prepare retention and promotional messages.
- Review every message before it is sent.
- Store communication history in the customer profile.

Management Dashboard:

- View daily and monthly sales and revenue.
- View popular frames and lenses.
- View low-stock items and customer activity.
- View OptoForecast insights and OptoRecall alerts.

Administrator / Optometrist Functions:

- Manage user accounts and access permissions.
- Create and review eye-examination and prescription records.
- Access all customer, prescription and purchase histories.
- Review LensIQ suggestions professionally.
- View sales, inventory and management reports.
- Use OptoForecast and OptoRecall.
- Manage system settings.

Optical Staff Functions:

- Register, search and update customer records.
- Manage frame, lens and accessory information.
- Record stock received and routine stock operations.
- Process sales, payments and receipts.
- Manage supplier information.
- Use LensIQ while assisting customers.
- Support approved customer follow-up.

6.2.3 User Experience Considerations

Simple Interface: Screens follow the order of the existing workflow so that the owner and staff can use the system with minimal training.

Consistent Layout: Bootstrap components provide a consistent and responsive layout across all screens.

Clear Feedback and Error Prevention: The system validates prescription values, prices and quantities and shows clear messages after each action (Nielsen, 1993).

Reliability: Core functions continue to work even when AI assistance is temporarily unavailable.

6.2.4 Core System Features Summary

Feature Category	Specific Features
User Access	Secure Login, Role-Based Access, User Management, Password Management
Customer Management	Customer Registration, Search, Customer Profile, Purchase History

Prescription Records	SPH, CYL, AXIS, ADD and PD Recording, Examination Notes, Prescription History
Product Management	Frame, Lens and Accessory Details, Prices, Suppliers
Inventory	Stock Quantities, Automatic Deduction, Stock Movement History, Low-Stock Alerts
Sales and Billing	Sales, Discounts, Payments, Balances, Receipts and Invoices
Suppliers	Supplier Details, Products Supplied, Purchase Records
Communication	Order-Ready Notifications, Review Reminders, Promotional Messages
Dashboard	Sales, Revenue, Popular Products, Low-Stock Items, AI Insights

Table 8: Core system features summary

6.3 LensIQ

6.3.1 Overview

LensIQ is an AI-assisted lens decision-support feature. It helps the owner and the staff member identify suitable lens products to discuss with a customer by combining the customer's SPH, CYL and ADD values with their lifestyle, preferences and budget and with the lenses currently available at Indian Optician. LensIQ recommends a suitable lens material, index, design, coating or catalog product, while the prescription itself is determined by the optometrist during the eye examination and stored by the Core Optical Center Management System. LensIQ brings a consistent approach to lens selection, which should consider the prescription, lens material, index, coating and the wearer's visual needs.

6.3.2 Core Functionality

Prescription Inputs:

- SPH for each eye.
- CYL for each eye.
- ADD, where prescribed.

Lifestyle Inputs:

- Occupation, which gives context about the customer's normal visual activities.
- Daily screen time, which helps LensIQ consider lens and coating options suited to long periods of digital screen use.
- Outdoor activity, which helps LensIQ consider outdoor-related options such as photochromic lenses where appropriate and available.
- Driving habits, which help LensIQ consider options such as anti-reflective coatings where relevant.

Customer Preferences:

- Budget, so that the recommendation balances suitability and affordability.
- Preference for thinner or lightweight lenses.
- Coating preferences, such as anti-reflective or photochromic options where available.

System-Supplied Inputs:

- Compatible available lens catalog options.
- Lens material, index and design/type.
- Available coatings.
- Price and stock/availability status.

Outputs:

- Ranked suitable lens options selected only from valid, available catalog products.
- Recommended material, index, design and coating, where applicable.
- A clear reason for each recommendation.
- A notice that professional review is required.

Use of Prescription Values:
SPH: Indicates the strength of the main spherical correction. A low prescription is often well served by a standard, cost-effective lens material, while a stronger prescription may make a thinner higher-index lens, such as 1.60 or 1.67, worth considering, subject to budget, preferences and catalog availability.

CYL: Indicates the amount of astigmatism correction and, together with SPH, is used to confirm that a candidate lens product supports the customer's prescription.

ADD: Indicates that near-vision power has been prescribed. Where ADD is present, LensIQ may consider reading, bifocal or progressive designs, selecting among them according to the customer's lifestyle, intended use, preferences, budget and available catalog.

Processing:
Stage 1 – Compatibility Filtering: The Node.js/Express backend retrieves the customer's SPH, CYL and ADD values for both eyes and the lens catalog from MySQL, and shortlists the lenses whose supported prescription ranges and current availability match the customer's needs.

Stage 2 – Gemini-Assisted Recommendation: The prescription values, shortlisted lenses, lifestyle inputs, budget and preferences are sent to Gemini, which ranks the shortlisted options and explains the reasons for each recommendation.

Processing Flow: Stored Prescription → SPH, CYL and ADD → Backend Compatibility and Availability Filtering → Compatible Lens Candidates + Lifestyle, Budget and Preferences → Gemini Ranking and Reasons → Backend Response Validation → Professional Review

6.3.3 Benefits

- Provides consistent lens-product recommendations for customers with similar

needs.

- Personalizes recommendations using lifestyle, budget and preferences.
- Recommends only lenses that are compatible with the prescription and available in stock.
- Shares only the information needed for lens selection with the AI service, supporting customer privacy.
- Helps the staff member explain lens options clearly to customers.
- Supports, rather than replaces, the professional judgment of the optometrist.

6.3.4 Technical Implementation

- Prescription values (SPH, CYL and ADD for each eye) retrieved from the MySQL database.
- Lens catalog specifications, including supported prescription ranges and stock status, retrieved from MySQL.
- Compatibility and availability filtering performed by the Node.js/Express backend.
- Structured context sent to the Gemini API through the backend, with the API key stored only on the server.
- Gemini ranking and explanation of the shortlisted lens options.
- Checking that every recommended lens ID exists, is available and was among the shortlisted options.
- Display of validated recommendations with reasons in the LensIQ screen for professional review.

6.3.5 Safety Considerations

- LensIQ does not conduct eye examinations or diagnose eye conditions.
- LensIQ does not create or change a prescription.
- LensIQ recommends only compatible lenses from the OptoPick catalog.
- Technical compatibility is checked by OptoPick’s catalog rules, not by Gemini.
- The final decision on lens suitability remains with the Administrator / Optometrist.

6.3.6 LensIQ Features Summary

Feature Category	Specific Features
Prescription Context	SPH, CYL, ADD where prescribed
Lifestyle & Preferences	Occupation, Screen Time, Outdoor Activity, Driving, Budget, Thickness/Weight and Coating Preferences
Catalog Compatibility	Lens Type/Design, Material, Index, Supported Prescription Range, Price, Stock/Availability
Recommendation	Suitable Compatible Lens Options, Material/Index/Design/Coating, Reasons
Validation	Compatibility Check, Catalog Check, Availability Check, AI Response Validation, Professional Review

Table 9: LensIQ features summary

6.4 OptoForecast

6.4.1 Overview

OptoForecast is an AI-assisted inventory demand feature. It helps the owner identify frames, lenses and accessories that may need attention because of rising demand, possible shortage, slow movement or possible overstock. Inventory management literature notes that purchasing without demand information leads to both shortages of popular items and excess stock of slow-moving items (Wild, 2017).

6.4.2 Core Functionality

Data Prepared by OptoPick:

- Units sold for each product over selected periods.
- Current stock quantity and low-stock limit.
- Recent sales trend and time since the last sale.
- Product and category totals.

AI Analysis Outputs:

- Demand Signal: Low, Stable, Increasing or High.
- Stock Risk: None, Possible Shortage or Possible Overstock.
- Recommended Action: Monitor, Review Reorder or Review Stock Reduction.
- Reason: a clear explanation of the insight.

6.4.3 Benefits

- Identifies popular products before they run out.
- Highlights slow-moving products that tie up capital.
- Supports better purchasing decisions based on sales data.
- Explains insights in simple language for the owner.

6.4.4 Technical Implementation

- Sales and stock summaries calculated by the OptoPick backend from the MySQL database.
- Structured summaries sent to the Gemini API through the backend.
- Checking that the AI response uses only the permitted categories and products.
- Display of insights in the management dashboard.

6.4.5 Considerations

- OptoPick calculates all sales and stock figures itself; the AI only interprets them.

- OptoForecast provides demand signals and recommendations rather than exact sales predictions.
- OptoForecast does not place orders automatically; the final purchasing decision remains with the owner.

6.4.6 OptoForecast Features Summary

Feature Category	Specific Features
Data Preparation	Sales by Period, Current Stock, Low-Stock Limit, Sales Trend
Demand Insight	Demand Signal, Stock Risk, Recommended Action, Reason
Management Support	Dashboard Insights, Owner Review

Table 10: OptoForecast features summary

6.5 OptoRecall

6.5.1 Overview

OptoRecall is an AI-assisted customer follow-up and retention feature. It identifies customers who may need attention, explains why, and prepares suitable follow-up messages for staff review. Retaining existing customers through organized recall is an important part of long-term success for optical practices.

6.5.2 Core Functionality

Follow-up Rules:

- Orders marked as ready for collection trigger order-ready notifications.
- Customers whose last eye examination exceeds the review period are marked for review reminders.
- Customers who have not purchased for a long period are marked as follow-up candidates.

Customer Analysis:

- Time since last purchase and last eye examination.
- Purchase frequency and previous purchases.
- Previous communication and response.

AI Outputs:

- Attention level for each customer.
- Reasons why follow-up may be useful.
- Recommended follow-up action.
- Editable message draft.

6.5.3 Benefits

- Reduces uncollected orders.
- Reminds customers when eye reviews may be due.
- Helps recover customers who have stopped visiting.
- Saves staff time when preparing messages.

6.5.4 Technical Implementation

- Follow-up rules applied by the OptoPick backend.
- Customer summaries prepared using customer IDs instead of names and telephone numbers.
- Gemini API integration through the Node.js backend.
- Display of attention levels, reasons and message drafts for staff review.
- Storage of sent messages in the customer's communication history.

6.5.5 Safety Considerations

- OptoRecall uses careful wording such as "may be due for a review".
- It never states that a customer definitely needs new glasses or a new prescription.
- Every message is reviewed and approved by staff before it is sent.

6.5.6 OptoRecall Features Summary

Feature Category	Specific Features
Follow-up Rules	Order-Ready Notifications, Review Reminders, Lapsed Customer Detection
Customer Insight	Attention Level, Reasons, Recommended Action
Communication	Editable Message Drafts, Staff Approval, Communication History

Table 11: OptoRecall features summary

6.6 Project Scope

The project scope defines the functions and deliverables that will be completed within the academic project period.

6.6.1 In Scope

The proposed system will include the following functionalities:

User Management

- Administrator / Optometrist and Optical Staff Login
- Role-Based Access Control

- User Account and Password Management

Customer and Prescription Management

- Customer Registration, Search and Profile Management
- Digital Eye-Examination and Prescription Records
- Chronological Prescription History
- Customer Purchase and Communication History

Product and Inventory Management

- Frame, Lens and Accessory Management
- Stock-In and Authorized Returns
- Automatic Sales-Linked Stock Deduction
- Stock Movement History
- Low-Stock and Out-of-Stock Alerts

Sales and Billing

- Sales Transactions
- Automatic Price, Discount and Balance Calculation
- Payment Recording
- Digital Receipts and Invoices

Supplier Management

- Supplier Details and Products Supplied
- Purchase Records and Supplier Prices

Customer Communication

- Order-Ready Notifications
- Eye-Review Reminders
- Retention and Promotional Messages with Staff Review

Reports and Dashboard

- Sales and Revenue Summary
- Product Performance
- Inventory and Low-Stock Summary
- Customer Activity

AI Features

- LensIQ – AI-Assisted Lens Recommendation
- OptoForecast – AI-Assisted Demand Insight
- OptoRecall – AI-Assisted Customer Follow-up
- Gemini API Integration through the Backend

Project Deliverables

- Fully functional OptoPick web application
- MySQL database for the system
- LensIQ, OptoForecast and OptoRecall features

- Test cases and testing results
- User guide for the owner and staff
- Final project report and presentation

6.6.2 Scope Justification

The scope is carefully defined to:

- **Focus on Core Problems:** Address the problems identified at the center.
- **Achievable Timeline:** Complete the system within the academic project period.
- **Resource Feasibility:** Match the available team skills and resources.
- **User Needs:** Prioritize the functions used daily by the owner and staff.
- **Future Expansion:** Allow additional features to be added in future versions.

6.7 Project Feasibility

6.7.1 Technical Feasibility

Technology Availability:

- React.js, Node.js, Express.js and MySQL are mature, free and well documented.
- The Gemini API provides official documentation and developer support.
- Team members have relevant web development skills.

Platform Compatibility:

- OptoPick runs in standard web browsers such as Google Chrome and Microsoft Edge.
- The system runs on the existing Windows computer at the center.

Integration Capability:

- The backend connects the user interface, the MySQL database and the Gemini API.
- Core functions continue to work if the AI service is unavailable.

Technical Risk Assessment:

- Low risk: well-established technologies.
- Mitigation: validation of AI results, regular testing and human review.

6.7.2 Economic Feasibility

Development Costs:

- All main development tools are free or open source.
- The Gemini API offers a free tier suitable for development and testing; limited usage costs may apply during operation.
- Hardware: existing laptops and the center’s computer are sufficient.
- Infrastructure: existing internet connection.

Expected Benefits:

- Reduced administrative time.
- Fewer billing and record errors.
- Better stock control and purchasing.
- Improved customer retention.

Cost-Benefit Analysis:

- Low development and running costs.
- Significant operational benefits.
- Positive return on investment for the center.

6.7.3 Operational Feasibility

User Acceptance:

- Simple interfaces designed around the current workflow (Nielsen, 1993).
- Training and a user guide provided.
- Gradual transition from paper and Excel records.

System Compatibility:

- Works with the existing computer and internet connection.
- Minimal hardware requirements.

Operational Impact:

- Reduces manual workload.
- Improves accuracy of records and billing.
- Improves customer service.
- Requires minimal change to the way the center operates.

User Adoption Potential:

- The owner actively supports digitalization of the center.
- Only two users need to be trained.

6.7.4 Schedule Feasibility

Project Timeline:

- 20 July 2026 to 18 October 2026.
- 13 weeks of development.

Task Distribution:

- Tasks distributed among five team members.
- Parallel frontend, backend and AI development.

Risk Management:

- Agile methodology for flexibility.
- Weekly progress monitoring.

Schedule Risk Assessment:

- Realistic timeline for an academic project.

- Focused scope ensures achievability.

6.8 Development Methodology

6.8.1 Methodology Selection

The proposed project follows the Agile Scrum software development methodology. Agile development delivers software in small increments and involves the client throughout the project (Sommerville, 2016), while Scrum organizes the work into short, time-boxed sprints (Schwaber, K & Sutherland, J, 2020). This methodology supports:

- Continuous development
- Frequent testing
- Regular feedback from the owner of the center
- Iterative improvement of the AI features
- Effective team collaboration

6.8.2 Scrum Framework

Scrum Roles:

- Product Owner: Owner of Indian Optician
- Scrum Master: Jamaldeen Mohammed Amar
- Development Team: All team members

Scrum Artifacts:

- Product Backlog: All project requirements.
- Sprint Backlog: Tasks for the current sprint.
- Increment: Working system features at the end of each sprint.

Scrum Events:

- Sprint Planning: Define sprint goals and tasks.
- Regular Team Progress Meetings: Review completed tasks, pending work and issues.
- Sprint Review: Demonstrate completed work to the client and supervisor.
- Sprint Retrospective: Discuss process improvements.

6.8.3 Development Phases

Phase 1: Requirement Gathering (20 July – 29 July)

- Investigate the client organization.
- Conduct interviews with the owner and staff member.
- Observe the current workflow.
- Document system requirements.

Phase 2: Requirement Analysis and Planning (27 July – 6 August)

- Validate requirements with the owner.

- Prioritize features and create the product backlog.
- Conduct the feasibility study.
- Prepare the WBS and project schedule.

Phase 3: System Design (5 August – 19 August)

- Create UI/UX designs.
- Design the database.
- Develop and validate a prototype with the client.

Phase 4: Core System Development (17 August – 13 September)

- Develop login, customer and prescription modules.
- Develop frame, lens, inventory and supplier modules.
- Develop sales, billing and the dashboard.

Phase 5: AI Feature Development (2 September – 27 September)

- Set up Gemini API integration.
- Develop LensIQ.
- Develop OptoForecast.
- Develop OptoRecall and customer communication.

Phase 6: Integration and Testing (24 September – 7 October)

- Integrate all modules.
- Conduct system and AI testing.
- Fix bugs.

Phase 7: User Evaluation and Improvements (5 October – 12 October)

- Conduct user acceptance testing with the owner and staff.
- Make final improvements.

Phase 8: Deployment and Presentation (12 October – 18 October)

- Complete documentation.
- Deploy the system at the center.
- Prepare and deliver the final presentation.

6.8.4 Methodology Benefits

Flexibility: Adapt to changing requirements and client feedback.
Quality: Continuous testing throughout development.
Visibility: Regular demonstrations to the client and supervisor.
Risk Management: Early identification and resolution of issues. **Team Collaboration:** Clear roles and responsibilities.

6.9 Work Breakdown Structure (WBS)

6.9.1 WBS Overview

The project activities are divided into manageable tasks to ensure systematic project development.

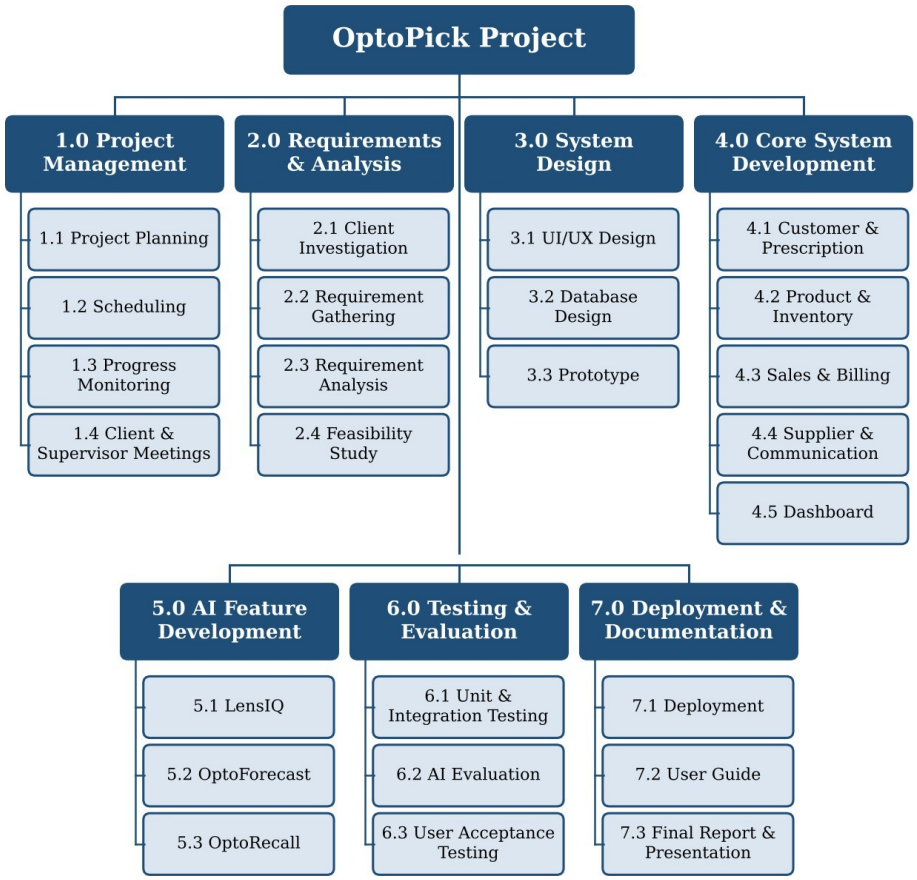


Figure 2: OptoPick Work Breakdown Structure

6.9.2 Task Dependencies

- Analysis depends on requirement gathering.
- Design depends on requirement analysis.
- Development depends on design.
- AI features depend on the core system data.
- Testing depends on development.
- Deployment depends on testing and user acceptance.
- Documentation is ongoing.

6.10 Task Allocation

6.10.1 Team Member Responsibilities

Team Member	Role	Responsibilities
Jamaldeen Mohammed Amar	Project Leader and AI Architect	Project Management, Client Liaison, Requirement Analysis, System Planning, Gemini API Integration, LensIQ, OptoForecast and OptoRecall Development, Final Review, Deployment Approval
Akbar Mohammed Akthas	Backend Lead and Database Engineer	Node.js and Express.js Backend Development, MySQL Database Design, Sales and Inventory Logic, Login and User Access, Backend Code Review, Deployment Support
Ranil Zibborah	Frontend Lead and UI/UX Designer	UI/UX Design, React.js Frontend Development, Bootstrap and CSS Styling, Dashboard and AI Feature Screens, Frontend Code Review
Mifla	Systems Analyst and Frontend Developer	Requirement Documentation, Customer, Prescription, Supplier and Communication Screens, Project Documentation, User Acceptance Testing Preparation
Afsal	Quality Assurance and Support Developer	Test Planning, Software Testing, AI Output Testing, Bug Tracking, User Guide, Presentation Support

Table 12: Team member responsibilities

6.10.2 Teamwork Approach

- **Regular Communication:** Weekly team meetings and progress updates.
- **Shared Responsibility:** Team members support each other.
- **Knowledge Sharing:** Sharing of technical knowledge on React.js, Node.js and the Gemini API.
- **Code Review:** Module leads review work before integration.
- **Collaborative Problem Solving:** Collective resolution of issues.

6.11 Project Schedule

6.11.1 Project Timeline Overview

The development of OptoPick is scheduled from 20 July 2026 to 18 October 2026 (13 weeks). The schedule follows the Agile Scrum methodology, with the work organized into sequential phases and sprints.

6.11.2 Gantt Chart

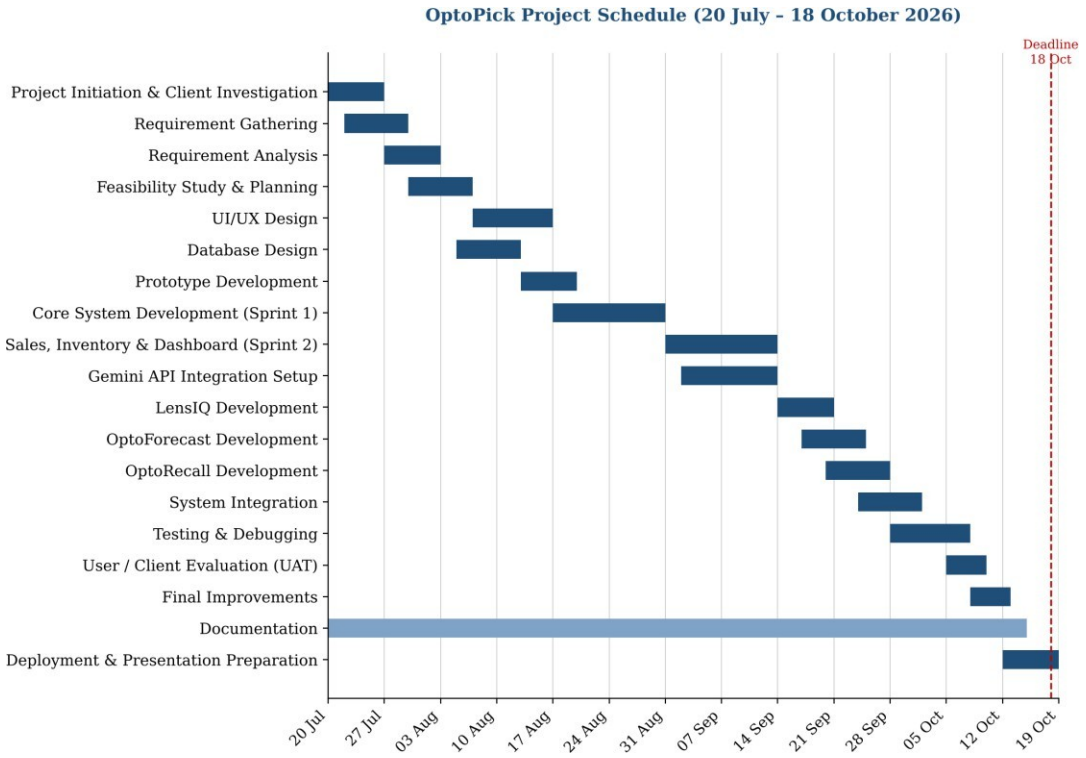


Figure 3: OptoPick Gantt Chart

6.11.3 Project Milestones

Milestone	Expected Completion Date	Status
Requirement Gathering Completed	29 July 2026	Planned
Requirement Analysis and Planning Completed	6 August 2026	Planned
UI/UX and Database Design Completed	16 August 2026	Planned
Prototype Validated with Client	19 August 2026	Planned
Core System Development Completed	13 September 2026	Planned
AI Feature Development Completed	27 September 2026	Planned
System Integration Completed	1 October 2026	Planned
System Testing Completed	7 October 2026	Planned
User Acceptance Testing Completed	9 October 2026	Planned
Final Deployment and Presentation	18 October 2026	Planned

Table 13: Project milestones

7. Resource Requirements

7.1 Software Requirements

The following software technologies and development tools will be used for the OptoPick project:

Frontend Development

- React.js
- Vanilla CSS
- Bootstrap

Backend Development

- Node.js
- Express.js

Database and Local Development

- MySQL
- XAMPP (including phpMyAdmin)

Artificial Intelligence

- Gemini API

The Gemini API will support the three AI features of OptoPick: LensIQ, OptoForecast and OptoRecall. It will be accessed only through the backend so that the API key and customer information remain protected.

Development and Testing Tools

- Visual Studio Code
- Postman
- Google Chrome and Microsoft Edge

Design Tools

- Stitch
- draw.io

Version Control

- Git
- GitHub

7.1.1 Purpose of Selected Technologies

Technology	Purpose in OptoPick
React.js	Builds the user interface from reusable components, suitable for screens such as customer profiles and dashboard (Meta Open Source, n.d).
Vanilla CSS and Bootstrap	Provide a consistent, responsive and professional appearance for all screens.
Node.js	Runs the backend using JavaScript, so the same language is used across the system (OpenJS Foundation, n.d).
Express.js	Handles requests, routes and access control for the backend (Express, n.d).
MySQL	Stores all OptoPick records and keeps sales and stock updates consistent (Oracle, n.d).
XAMPP	Provides a local MySQL server and phpMyAdmin for development and testing.
Gemini API	Provides AI reasoning for LensIQ, OptoForecast and OptoRecall .

Table 14: Purpose of selected technologies

7.2 Hardware Requirements

The OptoPick project requires basic development and testing hardware. It does not require specialized optical or medical hardware for the academic prototype. The following hardware resources will be used:

- Development laptops (Intel Core i5 or equivalent, 8 GB RAM, 256 GB SSD)
- Desktop or laptop computer at the center for running OptoPick
- Printer for receipts and invoices
- Internet connection for Gemini API access

- Wi-Fi network
- External drive or cloud storage for database backups

Core OptoPick functions operate on the center's computer, while an internet connection is required for the AI features.

7.3 Human Resources

7.3.1 Team Composition

Team Member	Role
Jamaldeen Mohammed Amar	Project Leader and AI Architect
Akbar Mohammed Akthas	Backend Lead and Database Engineer
Ranil Zibborah	Frontend Lead and UI/UX Designer
Mifla	Systems Analyst and Frontend Developer
Afsal	Quality Assurance and Support Developer

Table 15: Team composition

7.3.2 Required Skills

Project Leader and AI Architect Skills:

- Project Management
- Requirement Analysis
- System Planning
- Gemini API Integration
- Prompt Design

Backend Lead and Database Engineer Skills:

- Node.js and Express.js Development
- MySQL Database Design
- API Development
- Security and Access Control

Frontend Lead and UI/UX Designer Skills:

- React.js Development
- Bootstrap and CSS
- UI/UX Design

Systems Analyst and Frontend Developer Skills:

- Requirement Analysis
- Technical Writing
- React.js Development

Quality Assurance and Support Developer Skills:

- Test Planning
- Software Testing
- Quality Assurance
- Technical Writing

7.3.3 Team Development

Skill Development Plan:

- Team members with complementary skills.
- Knowledge sharing through collaboration.
- Learning new technologies together.
- Continuous professional development.

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